

Asad Ikram

MSc Data Analytics (Distinction) · Chevening Scholar 2024/25 · Applied machine learning and NLP, in production and in research
Lahore, Pakistan · +44 7482 376417 · asad.ikram53@gmail.com · linkedin.com/in/asad-ikram98 · asad-ikram-mc.github.io/portfolio · web-scraping-guide.com

EDUCATION

2024 - 2025 awarded Dec 2025	MSc Data Analytics, Distinction. University of Portsmouth, United Kingdom. Credit-weighted average 70.5/100 (Distinction threshold 70). Master's project 78/100. Supervisor: Dr Ella Haig. 180 UK credits, conventionally mapped to 90 ECTS. Modules included Intelligent Data and Text Analytics, Big Data and Machine Learning, Explainable AI, Data Management (80). Taught and assessed in English.
2017 - 2021 awarded Apr 2022	BSc Computer Science. National University of Computer and Emerging Sciences (FAST-NUCES), Pakistan. Four years full time, 130 credit hours completed. CGPA 2.75 on a 0.00 to 4.00 scale.

MACHINE LEARNING IN PRODUCTION

Feb 2025 - present	Founder and Chief Technology Officer , ArtemisAI Ltd (trading as ViralData), United Kingdom (remote). Running in production ahead of launch on 8 December 2026. • Six fine-tuned encoder classifiers for sentiment, emotion, toxicity, topic, language and intent, built from XLM-R, DistilBERT, DeBERTa-v3 and Detoxify bases and served from six SageMaker endpoints, called in parallel under 100ms. Custom per-sample loss weighted by label confidence, class balance and label source. • LLM labelling and distillation chain that produces the training data: Claude labels a stratified seed of about 6,000 posts together with its reasoning, the decision rules in that reasoning are mined into a prompt for DeepSeek, DeepSeek labels about 12,000 more, and any case below 0.80 confidence routes back to Claude. Result is about 18,000 machine-labelled items, which with synthetic cases makes a roughly 20,000-item confidence-weighted dataset, stratified 80/20. The models I fine-tune are encoder classifiers; I use Claude and DeepSeek through APIs and have not fine-tuned a generative model. • Deploy gate and retraining loop. A release ships only at macro-F1 above 0.85 on both a 97-post human holdout and a 4,000-item stratified test set, followed by a human check of 200 random posts. Weekly routing loop with three-tier confidence routing: roughly 75% auto-committed above 0.85, 18% sent to review between 0.60 and 0.85, 7% held below 0.60. Weekly KL-divergence drift alarms on output distributions and CI quality gates on every release. • Multimodal fusion classifier in production: a PyTorch network (1536 to 256 to 64 to 5, Adam, dropout 0.3, balanced class weights, best-validation checkpointing) over three concatenated 512-dimension CLIP embeddings of image, OCR text and caption, at macro-F1 0.93 on a hand-built gold set. Hybrid OCR pipeline at token-F1 0.928, Whisper speech transcription, and a networkx affinity graph applying network theory to a collaboration feature. • Data platform underneath the models: Meta Graph API ingestion of 30+ public pages daily into a layered lakehouse and Redshift warehouse, 111,000+ posts and 3.39 million comments processed. Products built on top: underperforming-post prediction, posting-time recommendation, drafting and crisis alerts. • <i>I also instrumented the measurement layer itself. The two annotation tiers agree on 82.9% of sentiment labels and 74.4% of emotion labels while the toxicity model reports 96.7% average confidence, and the dashboard's "accuracy" figure turned out to be inter-model agreement with no criterion behind it. Versions v1 to v3 were scored against the human holdout and v4 onward against Claude's labels, so part of the reported gain is the reference moving rather than the model improving. That moved my interest from building these instruments to asking what their outputs mean.</i>
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MACHINE LEARNING IN RESEARCH

Master's thesis: Cross-Platform Sentiment Analysis of Public Reaction to UK Economic Policies. Fine-tuned a BERT classifier to **macro-F1 0.878** against BiLSTM and lexicon baselines, with human validation at Cohen's $\kappa \approx 0.78$, temperature-scaling calibration of the predicted probabilities, McNemar and bootstrap significance testing of model differences, chi-square topic tests, LIME and integrated gradients for interpretability, NRC emotion analysis linking anger and fear to negative policy reactions, and Prophet forecasting of the daily sentiment index (MAE ≈ 0.02). Collected $\sim 279,000$ public posts on UK economic policy from X/Twitter, Reddit, YouTube and Quora via official APIs and licensed feeds, under ethics approval (Portsmouth TETHIC-2025-111094) and GDPR-compliant procedures. *The finding that shaped my research interest was not the accuracy: the same policy event produced structurally different sentiment and emotion profiles on different platforms, which meant the cross-platform aggregate rested on an untested assumption, that the classifier measures the same construct everywhere it reads.*

RESEARCH OUTPUT

Ikram, A. and Haig, E. "Is the Platform Part of the Measurement? A Protocol and Simulation Study for Cross-Platform Equivalence of Machine-Labelled Policy Sentiment." **Manuscript, complete; not published and not submitted.** Target venue ICWSM 2027, Round 3 (15 January 2027). Treats each social platform as a survey mode and applies generalizability theory, measurement invariance and differential item functioning, and score linking to machine-labelled text. In a 500-replication simulation with known ground truth, an index that pools platform-level scores at face value reports the wrong sign of the aggregate on roughly a quarter of days, purely from measurement differences between platforms.

DATA ENGINEERING AND SOFTWARE

Dec 2024 - present	Data Engineer, consultant, M+C Saatchi Fluency, United Kingdom (remote) - Operate the multi-platform social listening pipeline behind campaign intelligence for UK Government departments and for Amazon, Ford, Nike and Reckitt: Sprinkl and Brandwatch feeds plus custom collectors into AWS (ECS Fargate, Step Functions, Glue, Athena), with SageMaker NLP sentiment models on the output. 68 collectors ingesting ten platforms daily into a governed 156-column master table; 6.7 million items in a six-week window at 99%+ uptime. Automated repair agent as a named CI workflow: when a source changes shape the system diagnoses the break, regenerates the extraction logic and tests it against live pages before shipping. - Anti-bot handling, proxy rotation and TLS fingerprint management; weekly KL-divergence drift alarms on pipeline output distributions; documented filter rules, removal lists and pre-delete backups on government-facing datasets.
Feb 2022 - Aug 2024	Senior Data Engineer, Dubizzle Labs (Dubizzle, Bayut, Zameen, OLX), Pakistan Led a three-engineer team owning 500+ regional collectors across 15 MENA countries on Kubernetes/EKS, gathering multilingual platform data in English, Arabic, Urdu and Hindi. Redesigned the layered warehouse (RDL, ODL, ADL), improving ETL/ELT throughput by 60%. Promoted twice in two years.
2018 - present	Earlier and concurrent engagements Fix.com, Canada, consultant, May 2023 to present (part time): Scrapy/Scrapyd pipelines processing 50M+ data points from 200+ sources; Great Expectations validation with automated schema-drift detection, cutting bad production data by 70%. VendueTech, Ireland, Lead Data Engineer, Jun to Jul 2025: led three engineers on real-time auction ingestion, migrated Azure to AWS. Prefe, Italy, consultant, Sep to Dec 2024: rebuilt legacy collectors, cutting failure rate 60%. CXG, Dubai, Dec 2025 to Jan 2026: production collectors for luxury retail at 99%+ completeness against advanced anti-bot protections. Freelance data engineer (Level 2 seller), Dec 2018 to Feb 2026: 300+ five-star projects across 400+ websites for 50+ international clients.

METHODS AND TOOLS

Machine learning	Transformer fine-tuning (BERT, XLM-R, DistilBERT, DeBERTa-v3, Detoxify); LLM-assisted labelling and distillation into smaller encoder models; confidence-weighted loss and class-imbalance handling; CLIP multimodal embeddings and fusion networks; Whisper transcription; OCR pipelines; BiLSTM; PyTorch; SageMaker training, endpoints and weekly retraining loops.
Evaluation	Macro-F1 against human holdouts and stratified test sets; inter-coder reliability (Cohen's κ); temperature-scaling calibration; McNemar and bootstrap testing; chi-square; KL-divergence drift detection; LIME and integrated gradients; Prophet forecasting.
Measurement	Generalizability theory, measurement invariance and differential item functioning, score linking; treating annotators and platforms as measurement instruments rather than as ground truth.
Data collection	Scrapy, Playwright, Selenium; official platform APIs (Meta Graph, Reddit, YouTube); licensed feeds; proxy and anti-blocking management; TLS fingerprinting; detection anatomy; agentic browsers.
Engineering	Python, PySpark, SQL; AWS (SageMaker, Fargate, Glue, Athena, Redshift, Step Functions, Lambda, DynamoDB); Airflow; Docker and Kubernetes; PostgreSQL; CI/CD with automated quality gates. R functional rather than fluent.

AWARDS, PUBLIC WORK AND LANGUAGES

Chevening	Chevening Scholar 2024/25, the UK Foreign, Commonwealth and Development Office's flagship global scholarship, selected from over 70,000 applicants across 160 countries and awarded to fund the MSc at Portsmouth.
Public work	Author and maintainer of web-scraping-guide.com , a free public engineering reference on web data collection: 24 sections covering 60+ libraries and tools, six anti-bot systems, detection anatomy, agentic browsers, legal and ethical constraints, production architecture and data quality.
Languages	Urdu (native). English: MSc taught and assessed in English, IELTS Academic 6.5 overall, March 2024. No Swedish; I would begin learning on arrival.
Teaching	No formal university teaching. Mentored two junior engineers at Prefe and led three-engineer teams at Dubizzle Labs and VendueTech; wrote and maintain a public technical reference used by practitioners.